

# Smart Metro Ticket Booking System

## Requirement Analysis Phase

|               |                                   |
|---------------|-----------------------------------|
| Date          |                                   |
| Team ID       |                                   |
| Project Name  | Smart Metro Ticket Booking System |
| Maximum Marks | 4 Marks                           |

### 4. Technology Stack

This document sets out the technology stack proposed for the ServiceNow-based Metro Ticket Booking system, spanning the presentation, application logic, integration, data, notification, and security layers.

Technology Stack — Layered Architecture



Figure — Layered technology architecture of the ticketing solution

#### 4.1 Layer-wise Technology Breakdown

| Layer                     | Technology / Component                                                                          | Purpose                                                                                    |
|---------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Presentation              | ServiceNow Service Portal, UI Pages, WhatsApp Business Chatbot                                  | Passenger-facing booking form, fare preview, ticket display, and chat-based booking.       |
| Application / Process     | Service Catalog, Flow Designer, Business Rules, Script Includes, Catalog Client Scripts         | Captures booking input, orchestrates the booking-to-ticket workflow, executes fare logic.  |
| Integration               | Integration Hub, REST/SOAP APIs, Payment Gateway (UPI/Card/Wallet), QR Code Generation Library  | Connects the platform to external payment and QR services with logged, traceable calls.    |
| Data                      | ServiceNow Custom Tables — Station Master, Fare Rule Table, Ticket/QR Records, Transaction Logs | Stores the reference and transactional data needed for booking, validation, and reporting. |
| Notification              | Email Notifications, SMS Gateway, Push Notifications, WhatsApp Business API                     | Delivers QR tickets, payment confirmations, and exception alerts to passengers and staff.  |
| Reporting & Analytics     | Performance Analytics, Reports, Dashboards                                                      | Surfaces booking volume, revenue split, and peak-time insights for Metro Operations.       |
| Security & Administration | Access Control Lists (ACLs), OAuth 2.0, Data Encryption, Role-based Views                       | Restricts data and functionality by persona and secures payment/QR data end-to-end.        |

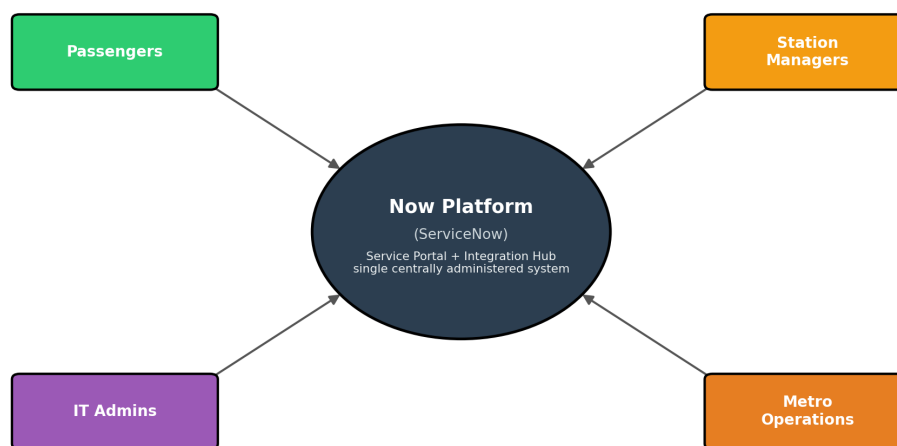
#### 4.2 Platform Rationale

- The Service Catalog offers a governed, reusable form framework for capturing structured booking input (Source, Destination, Passenger Type, Number of Tickets) with built-in client-side validation.
- Flow Designer allows the multi-step booking-to-ticket process to be orchestrated visually and with little code, making the workflow easier to audit and change without heavy scripting.
- Integration Hub standardizes outbound calls to the payment gateway and QR generation service, giving IT Admins centralized logging, retry handling, and credential management.
- Native Notifications keep ticket delivery inside the ServiceNow platform, avoiding a separate messaging system while still covering Email, SMS, Push, and WhatsApp channels.
- Performance Analytics and Reports reuse existing platform capabilities to give Metro Operations the data-driven insight it needs for route and peak-time planning, without adding a separate BI tool.

### 4.3 Environment Considerations

The solution is built to run entirely on the ServiceNow platform (Now Platform), using out-of-the-box Service Portal capabilities for the front end and Integration Hub spokes for all third-party connections. This keeps the technology footprint small, avoids the licensing overhead of a separate mobile app in this initial phase, and lets every persona — Passengers, Station Managers, Metro Operations, and IT Admins — be served from one centrally administered system.

#### Single-Platform Deployment Model



*No separate mobile app or external BI/messaging system required in this phase*

*Figure — Single-platform deployment model serving all four personas*